

■ XenChat ■

PRODUCT REQUIREMENTS DOCUMENT

v 2 . 0

Privacy-first · Game-Vibed · Stealth Messaging

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1. PRODUCT VISION & PHILOSOPHY

XenChat is a React Native messaging app that fuses the privacy of Signal with the visual identity of a hacker-RPG game. Every screen, animation, and color communicates stealth, control, and power.

◆ Core Principles

- Privacy should feel invisible – never a burden, never obviously labeled.
- Messages in Privacy Mode look like the sender is just typing weird. No lock badge.
- Reveal is an earned action: triple-tap a message, enter your code – only then real text surfaces.
- Auto-re-encryption after 2 min of inactivity. No 'LOCKED' banner. It happens silently.
- The UI speaks game, not app – neon grid, cipher blocks, glitch animations every screen.

■ **Design rule: If you can tell XenChat is a 'private messaging app' from the lock screen, the design has failed.**

◆ What Changes From v1.0

v1.0 Behaviour	v2.0 Behaviour
Flutter	React Native (Expo)
Unlock once → 10 min session	Triple-tap per message → 2 min auto re-encrypt
'Private Chat Enabled' banner shown	Zero visible lock indicator anywhere
Cipher block (#H8K2X7) displayed	Free: cipher block Pro: AI Ghost Language
No games tab	Dedicated Games Hub – center nav tab
No bottom nav bar	WhatsApp-style 5-tab nav: Chats · Groups · Games · Profile · Settings
No onboarding spec	Full Instagram/Google-style sign-in + username flow

■ 2. TARGET AUDIENCE & PERSONAS

Persona	Age	Why XenChat	Key Feature
The Cautious Texter	18-24	Shared devices at home / hostel	Triple-tap reveal + auto re-encrypt
The Gamer Socialiser	16-28	Wants chat that feels like a game	Games Hub + neon UI
The Privacy Nerd	22-35	Knows what E2EE means	Vault + Panic Mode + libsignal
The Gossip Group	18-30	Sensitive group drama	Incognito contacts + stealth notifs
The Couple	20-32	Private convos on shared phone	Secret Vault + self-destruct msgs

■ 3. SECURITY MODEL

◆ 3.1 Chat States

Every conversation exists in exactly one state at any time:

State	What the Viewer Sees	How to Enter
Normal Mode	Real messages, always visible	Default for all new chats
Privacy Mode - Locked (Free)	Cipher blocks e.g. #1K2DR	User enables Privacy Mode
Privacy Mode - Locked (Pro)	AI Ghost Language scramble	User enables Privacy Mode + Pro sub
Privacy Mode - Revealed	Real messages for 2 min window	Triple-tap → passcode / biometric
Vault - Locked	Not shown in main chat list at all	Open Vault section
Panic Mode	Decoy chat list with fake convos	Enter duress PIN anywhere in app

◆ 3.2 The Reveal Mechanic

v2.0 replaces the v1.0 single 10-minute session unlock with a per-message-cluster reveal + silent auto re-encrypt:

- User triple-taps any message bubble inside a Privacy Mode chat.
 - A minimal dark modal rises: 4-char code input, fingerprint icon, or face-ID prompt. No 'LOCKED' text.
 - On success: visible messages reveal real text. A faint countdown glyph (■) appears at thread top.
 - After 2 minutes of inactivity (no taps, no scrolling), messages silently revert to scrambled / cipher form.
 - If user is actively scrolling, the timer resets on each scroll. Only true inactivity triggers re-encrypt.
- No visible lock label anywhere – not in chat list, not in the header, not as a badge. The scrambled display IS the lock.

◆ 3.3 Passcode Options

- Fingerprint biometric (primary, fastest)
- Face unlock (Android only – documented as best-effort)
- 4-character alphanumeric code – attempt-limited: 5 wrong attempts → 30s lockout, exponential backoff

◆ 3.4 Secret Vault Security

- Requires fingerprint + PIN simultaneously. Single factor is not sufficient.
- Vault re-locks the instant the app backgrounds. The 2-min window does NOT apply.
- Vault chats never appear in the normal Chats tab, even in scrambled form.

◆ 3.5 Panic Mode

- User pre-configures a duress PIN (e.g. 0000) in Settings > Advanced Security.

- Entering the duress PIN at any unlock prompt loads the decoy chat list.
- Real data is in a separate encrypted partition – not mounted, not just hidden at UI layer.
- Decoy chats are pre-configured by the user (fake names, fake message history).
- Forensic risk: decoy partition must be physically separate storage. Tested against device inspection simulation before shipping.

■ 4. FEATURE SPECIFICATIONS

◆ 4.1 Stealth Message Display (Privacy Mode Visual)

When a chat is in Privacy Mode and locked, messages are NOT plaintext. The display depends on the user's subscription tier:

Tier	What the viewer sees (locked state)
Free	Cipher blocks – #1K2DR, #X9PT4, #3MZ7Q – one block per message, derived from message hash (consistent, not random)
Pro / Pro+	AI Ghost Language – same sender tone/vocab, completely scrambled meaning. Reads like a half-typed real message.

Real Message	Free Display (locked)	Pro Display (locked)
Meet me at 8 PM at the station	#4RZ9M	"bro station PM at me the 8 meet"
Did you tell her about us?	#7KP2X	"you her tell about did us? lol"
Delete this after reading	#9QT1W	"after this reading delete ok bro"

■ **Design goal: a glancing observer thinks the sender typed something weird (Pro) or the preview failed to load (Free) – not that the message is 'protected'.**

◆ 4.2 Triple-Tap Reveal + Auto Re-Encrypt

Reveal Flow

- User triple-taps any message bubble.
- Dark overlay modal rises from bottom: biometric prompt or 4-char code field. No label saying 'locked'.
- On success: all visible thread messages reveal real text. Glitch scanline animation plays (<0.5s).
- Faint glyph '■ 2:00' appears at thread top, counting down. Not prominently visible.

Auto Re-Encrypt

- After 2 minutes of no interaction, messages silently re-scramble to cipher/Ghost form.
- Quick reverse-glitch animation (0.3s). No notification, no modal, no sound.
- Timer resets on any scroll or tap event – only pure inactivity triggers it.

◆ 4.3 End-to-End Encryption

- Signal-protocol-style key exchange per conversation (X3DH + Double Ratchet).
- Library: libsignal-protocol-javascript – no custom crypto.
- Server stores and relays ciphertext only. Cannot read message content.
- Private keys never leave the device unencrypted.
- Media encrypted client-side before upload to S3-compatible object storage.

◆ 4.4 Incognito Contacts

- Privacy Mode contacts in the chat list show as Player #XXXX (random 4-digit ID) + no avatar.
- Real name and avatar appear only after triple-tap reveal succeeds.

- After the 2-min session window closes, chat list label reverts to Player #XXXX.
- Chat list preview shows cipher block or Ghost Language snippet – not real text.

◆ 4.5 Self-Destruct Messages

- Sender sets a timer per message: 10s / 1min / 1hr / 24hr.
- Timer starts when the receiver opens and views the message – not from send time.
- On expiry: message deleted from both devices and server. Replaced with '– destroyed –' marker.
- Self-destruct messages always show in cipher / Ghost form in the chat list preview.

◆ 4.6 Screenshot Detection & Stealth Alert

- Android: detect screenshot event; show a subtle in-chat system event line – not a push notification.
- iOS: best-effort only – API limitations documented. Generic privacy reminder shown instead.
- Alert text inside chat thread: '■ Someone captured the screen.' – not in notification shade.

◆ 4.7 Secret Vault

- Accessible via a dedicated Vault section – not the main Chats tab.
- Unlock requires fingerprint + PIN simultaneously.
- Re-locks instantly on app background – no 2-min grace period.
- Vault chats do not appear in main Chats tab at all.
- Vault has its own encrypted data partition separate from normal chats.

◆ 4.8 Panic Mode

- User pre-sets a duress PIN in Settings > Advanced Security (hidden section).
- Entering duress PIN anywhere in app instantly loads a decoy chat list.
- Real data in a separate encrypted partition – not loaded, not just UI-hidden.
- Decoy chats pre-configurable (fake names, fake message history).
- Exiting Panic Mode requires real device biometric – duress PIN alone cannot exit.

◆ 4.9 AI Ghost Language (Privacy Mode Transformer) · PRO / PRO+ ONLY

Paid feature. Free tier users see cipher blocks (#1K2DR) instead. AI Ghost Language is never hinted at in the UI – it just looks like the Pro user's messages are oddly worded.

Free Tier – Cipher Block Display

- Locked messages show as alphanumeric cipher strings: #1K2DR, #X9PT4, #3MZ7Q.
- Format: # + 5 uppercase alphanumeric chars, derived from message hash (consistent on re-render).
- Triple-tap + passcode still works identically to reveal real text.
- No inline 'Upgrade' prompt inside chat – upsell only in Settings > Privacy.

Pro / Pro+ Tier – AI Ghost Language

- At send time, server calls LLM API. Instruction: 'Rewrite using sender's own vocabulary. Scramble the meaning. Same casual tone. Output rewritten text only.'

- AI-transformed version stored as display string alongside encrypted original.
- Triple-tap reveal shows the stored original directly – AI never decodes.
- Fallback if API unavailable: cipher block shown automatically. Message sends normally.
- Pro+: extra transform styles – Poetic, Formal, L33tspeak – selectable per chat.

Feature	Free	Pro	Pro+
Privacy Mode	✓	✓	✓
Triple-tap reveal	✓	✓	✓
Locked display style	#1K2DR cipher block	AI Ghost Language	AI Ghost Language
Transform styles	–	Default only	Default + Poetic + Formal + L33t
AI fallback	Cipher block	Cipher block	Cipher block
Ghost Language in Vault	–	–	✓

◆ 4.10 Anonymous Rooms

- Temporary group chats identified by a 5-char Room ID (e.g. X7A91).
- Users join with the ID – no name required, shown as Agent #XXXX.
- Room and all message history permanently deleted 30 min after the last member exits.
- Room creator can set a max participant count (2-20).
- Not in the main chat list – accessed via Games Hub > Drop Zone.

◆ 4.11 Standard Messaging Features

- 1-to-1 and group text chat, voice notes, voice and video calls.
- Photo, video, document sharing – encrypted before upload.
- Status / Stories – visible only to contacts, ephemeral 24hr.
- Stickers, GIFs, emoji reactions.
- Read receipts and typing indicators – off by default in Privacy Mode chats.
- Multi-device login and sync. Encrypted cloud backup.
- Dark game-themed UI (default), custom neon chat themes.

◆ 4.12 Notification Stealth

- Push notifications show only: 'New Incoming Signal' – no sender name, no preview.
- User can optionally allow sender name for Normal Mode chats only.
- Badge count: shown. Chat preview in notification shade: never.

■ 5. NAVIGATION & UI ARCHITECTURE

◆ 5.1 Bottom Navigation Bar

Five tabs – styled as glowing HUD icons on a dark game interface:

Tab	Icon Style	Content
CHATS	Speech bubble ■	All 1-to-1 conversations
GROUPS	Three nodes ■	Group chats and broadcast channels
GAMES	Controller ■ (CENTER)	Games Hub – 5 unique XenChat games
PROFILE	Hexagon avatar ■	Username, bio, avatar, stats
SETTINGS	Terminal gear ■	Privacy, Vault, Panic Mode, themes, subscription

■ **GAMES is center position – prime real estate – reinforcing that XenChat is a platform, not just a messenger.**

◆ 5.2 Onboarding (Instagram-Style)

- Splash screen with glitch animation of the XenChat logo.
- Sign Up: Phone number (primary) or Google OAuth.
- OTP verification (phone) or Google account confirm.
- Username selection – unique @handle style.
- Avatar upload or auto-generated pixel-art avatar.
- Contact sync permission prompt.
- Tutorial overlay: 'Triple-tap any message. You'll see.' – gamified, minimal.

■ 6. UNIQUE IN-APP GAMES (Games Hub)

Every game is thematically tied to XenChat's privacy/hacker identity – not generic mini-games. Playable inside a chat or as standalone sessions from the Games tab.

◆ Game 1 – CIPHER HUNT (2-player, async)

- Each player sends a word. XenChat encodes both into cipher strings.
- Each player must guess the other's original word within 3 attempts.
- Scoring: 3 pts for 1st attempt, 2 pts for 2nd, 1 pt for 3rd, 0 for fail.
- Best of 5 rounds. Streak badge displayed on profile.
- Unique mechanic: uses the same cipher algorithm as Privacy Mode – players learn to read XenChat cipher through gameplay.

◆ Game 2 – GHOST WRITER (2-player, turn-based)

- One player writes a sentence. XenChat AI rewrites it in Ghost Language style.
- The other player reads the scrambled version and must reconstruct the original meaning.
- Semantic similarity > 80% = win. 5 rounds, alternating sender/guesser.
- Playing Ghost Writer trains players to read Privacy Mode messages – core loop.

◆ Game 3 – DEAD DROP (multiplayer strategy, 4-8 players)

- Players assigned roles: Spy, Double Agent, Handler, Analyst.
- A secret 'package' (random emoji sequence) is passed player-to-player via Anonymous Room messages.
- The Analyst must intercept and identify the package before it reaches the Handler.
- Spy and Handler must disguise the package using Ghost Language transforms.
- Async play – 24-hour round timer. Push notification on your turn.

◆ Game 4 – NEON DUEL (1v1, real-time reaction)

- Two players in a chat tap a glowing neon target the instant it appears.
- Target timing randomised, target moves – not static. Best of 7.
- Winner gets a glowing WINNER badge in the chat header for 24h.
- Real-time via WebSocket. Latency shown post-game: 'You won by 34ms'.

◆ Game 5 – MEMORY VAULT (solo, daily challenge)

- Player shown a sequence of cipher-style symbols for 5 seconds. Must recreate from memory.
- Difficulty increases daily. Global leaderboard (anonymous handles only).
- Completing daily challenge unlocks cosmetic rewards – chat themes, sticker packs.
- Only solo game – drives daily opens via streak mechanic.

Game	Players	Mode	Unique Hook
Cipher Hunt	2	Async turn-based	Uses XenChat's own cipher – learn Privacy Mode through play
Ghost Writer	2	Async turn-based	Train yourself to decode AI Ghost Language messages

Game	Players	Mode	Unique Hook
Dead Drop	4-8	Async strategy	Multiplayer espionage using real Anonymous Rooms
Neon Duel	2	Real-time reflex	WebSocket 1v1, millisecond precision, 24h badge reward
Memory Vault	Solo	Daily challenge	Streak + global leaderboard + cosmetic unlocks

■ 7. SYSTEM FLOW

Step	Action	Result
1	App open → sign in via phone OTP or Google OAuth	Authenticated; contact list synced
2	Chat list loads	Privacy Mode contacts → Player #XXXX + cipher/Ghost preview; Normal → real name
3	User opens Privacy Mode chat	Thread visible – all messages in cipher block or Ghost Language form
4	User triple-taps a message	Minimal dark modal: biometric or 4-char code prompt – no 'LOCKED' label
5	Unlock success	Glitch-decode animation; real messages visible for 2-min window
6	2 min inactivity passes	Silent re-scramble – no banner, no sound, no alert
7	App backgrounds	All Privacy Mode chats re-lock instantly; Vault always re-locks
8 ALT	Duress PIN entered	Decoy chat list loads; real partition unmounted
9 ALT	Screenshot detected (Android)	Subtle in-chat line: '■ Someone captured the screen.'
10 ALT	Self-destruct timer expires	Message deleted both sides; '– destroyed –' marker left

■ 8. TECHNICAL ARCHITECTURE

◆ 8.1 Frontend

- Framework: React Native (Expo managed workflow) – single codebase, Android + iOS.
- Navigation: React Navigation v7 – Bottom Tab Navigator (5 tabs) + Stack Navigator per tab.
- State: Zustand (global) + React Query (server data) + MMKV (local encrypted storage).
- Animations: Reanimated 3 + Skia for glitch/decode effects; Lottie for splash and loading.
- Crypto on device: react-native-libsodium for key management.
- Biometrics: react-native-biometrics – fingerprint + face ID unified API.

◆ 8.2 Backend

- Runtime: Node.js (Express) – REST + GraphQL hybrid.
- Real-time: Socket.io WebSocket for messages, typing, presence, Neon Duel game state.
- Auth: Google OAuth 2.0 (primary) + phone OTP via Firebase Auth (backup).
- AI: Server-side LLM API call at send time for Ghost Language transform. Stored as display string alongside encrypted original.

◆ 8.3 Database & Storage

- PostgreSQL: User accounts, contacts, metadata, game scores, leaderboards.
- Messages: Encrypted blobs only – server never stores or sees plaintext.
- Media: S3-compatible object storage (Cloudflare R2). Encrypted client-side before upload.
- Redis: Session tokens, Socket.io room state, Anonymous Room lifecycle (TTL).

◆ 8.4 Encryption Layer

- Signal-protocol-style per-conversation session keys: X3DH + Double Ratchet.
- Library: libsignal-protocol-javascript – vetted, audited, no custom crypto.
- Private keys in device secure enclave / Keystore. Never transmitted.
- Server sees: public keys, ciphertext, encrypted media – nothing else.

◆ 8.5 Games Infrastructure

- Cipher Hunt / Ghost Writer / Dead Drop: Async – turn state in PostgreSQL; push notifs via FCM/APNs.
- Neon Duel: Real-time WebSocket room per match; server authoritative for tap timestamps.
- Memory Vault: Client-side logic; score submitted on completion; daily seed from server.

◆ 8.6 Anonymous Rooms

- Short-lived Socket.io rooms keyed by Room ID.
- Room state + history in Redis with TTL: 30 min after last participant exits.
- On expiry: Redis key deleted + PostgreSQL log purged. No recovery possible.

■ 9. BUILD ROADMAP

Phase	Features	Goal
Phase 1 MVP	Google OAuth + phone OTP onboarding Core 1-to-1 chat + E2EE (libsignal) Privacy Mode + cipher block display (Free tier) Triple-tap reveal + 2-min auto re-encrypt Incognito contacts (Player #XXXX) 5-tab bottom nav bar Stealth push notifications	Working private chat app
Phase 2 Depth	Secret Vault (fingerprint + PIN) Panic Mode (decoy partition) Anonymous Rooms Self-Destruct Messages Screenshot detection (Android) Games tab + Neon Duel + Memory Vault	Privacy depth + gamification
Phase 3 Monetize	AI Ghost Language – LLM integration (Pro/Pro+) Cipher Hunt + Ghost Writer + Dead Drop games Subscription billing – Razorpay + Stripe Daily streaks, leaderboard, cosmetic unlocks Voice modulator + extra transform styles UPI payments integration	Monetize + daily engagement

■ 10. MONETIZATION MODEL

Tier	Price	Includes
XenChat Free	■0 / month	Full E2EE · Privacy Mode · Triple-tap reveal · Cipher block display (#1K2DR) · 2 games (Neon Duel, Memory Vault)
XenChat Pro	■99 / month	Everything in Free · AI Ghost Language (unlimited) · All 5 games · Extra themes · Voice modulator · Priority support
XenChat Pro+	■199 / month	Everything in Pro · Ghost Language in Vault · Poetic / Formal / L33t transform styles · Secret Vault premium · Panic Mode · Early access

■ No ads – ever. Monetization is subscription-only, consistent with the privacy-first positioning.

- Razorpay for India – UPI, cards, net banking.
- Stripe for international.
- Annual plans at ~2 months free.

■ 11. RISKS & MITIGATIONS

Risk	Severity	Mitigation
4-char code brute-forceable	HIGH	Attempt limiting: 5 wrong → 30s lockout, exponential backoff. Failed attempts logged.
iOS screenshot detection unreliable	MED	Document as best-effort. Alert on Android only. iOS shows generic privacy note.
AI API unavailable at send time	MED	Auto-fallback to cipher block. Message sends without transform if API times out in < 2s.
Panic Mode decoy not forensically solid	HIGH	Real data in separate encrypted partition – not a UI toggle. Tested with device inspection simulation before shipping.
Anonymous Rooms used for abuse	MED	In-app report mechanism. Room creator can eject participants. Auto-delete limits window.
libsignal JS perf on low-end Android	LOW	Benchmark on Snapdragon 665 class. Key generation offloaded to background thread.
User forgets 4-char code + no biometric	MED	Recovery via OTP + device re-key. Privacy Mode chat history lost on re-key – by design, no server key escrow.

■ 12. GLOSSARY

Term	Definition
Privacy Mode	Per-chat setting that scrambles messages and enables triple-tap reveal.
Ghost Language	AI-transformed display text – same sender tone, scrambled meaning. The visual lock. Pro/Pro+ only.
Cipher Block	Free-tier locked display: #1K2DR style – hash-derived, consistent per message.
Triple-Tap Reveal	The gesture that triggers the passcode / biometric prompt to reveal real messages.
Auto Re-Encrypt	Silent reversion to scrambled state after 2 min of inactivity. No banner, no sound.
Incognito Contact	Privacy Mode contact shown as Player #XXXX in chat list until revealed.
Secret Vault	Extra-secure chat space. Requires fingerprint + PIN. Locks instantly on app background.
Panic Mode	Duress PIN that loads a decoy chat list while hiding the real data partition.
Dead Drop	The 4-8 player espionage strategy game in the Games Hub.
Anonymous Room	Temp group chat by Room ID. Auto-deleted 30 min after last user exits.
E2EE	End-to-end encryption. Only sender and receiver devices hold keys. Server sees ciphertext only.
Session Window	The 2-min inactivity window after triple-tap reveal before re-scramble occurs.

END OF DOCUMENT

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